

Heat-related mortality risk across Canada's neighbourhoods: a case time series analysis of 41 census metropolitan areas

[Author line]

Abstract

Extreme heat kills in Canada far beyond the deaths certified as heat, and its risk concentrates in particular neighbourhoods. A neighbourhood records too few deaths to estimate its risk directly, and national estimates have therefore resolved to the city and above. We applied a two-stage small-area design to the [XX] dissemination areas of Canada's 41 census metropolitan areas. Case time series fits within each city and age band were pooled in a multivariate meta-regression on composite vulnerability components and downscaled to an exposure-response curve for every dissemination area. Heat-related mortality risk varied [more within cities than between them], with relative risk at the 99th temperature percentile ranging from [XX] to [XX] across dissemination areas and [XX] excess deaths (95% empirical CI [XX–XX]) attributable to heat each year. [Vulnerability components / the dominant modifier] accounted for [XX]% of the between-area variation in heat risk. The analysis yields a national map of heat-related mortality risk at neighbourhood resolution, with minimum-mortality temperature and attributable burden for every dissemination area. We recommend the dissemination-area estimates as the input layer for municipal heat-response planning.

1 Introduction

Non-optimum ambient temperature accounts for 7.71% of mortality across 384 locations in 13 countries, Canada among them [?], and more than a third of warm-season heat-related deaths worldwide are attributable to human-induced warming [?]. In Canada, roughly 670 excess deaths over 2000 to 2020 have been attributed to extreme heat events across the country's twelve largest cities [?]. The June 2021 heat dome was associated with [619] deaths in British Columbia BC Coroners heat dome death review—verify count and dates against presentation. Province-wide assessments place a burden of comparable scale within single provinces [?]. Heat risk is not

spread evenly—it concentrates in particular places and populations, and area-level characteristics account for much of the difference. Across Europe, small-area studies find heat mortality higher where population density is greater, green space scarcer, and air pollution higher [? ?]. The same characteristics recur across cities and countries, with only their local weights differing [?].

Canadian heat-mortality work has operated above the neighbourhood scale. National modelling has estimated temperature–mortality associations across 111 health regions [?], multi-city work across the twelve largest cities [?], and provincial analyses within single provinces [? ?]. Neighbourhood-scale work has been single-city: Toronto mapped through composite vulnerability indices [?], Quebec’s dissemination areas through aggregate counts [?], and three cities through the national death registry [?].

Estimation below the city is hard because the unit records too few deaths to estimate a stable exposure-response curve of its own. Two routes get past it. Single-stage spatial models fit all areas at once, borrowing strength between neighbours through a conditional autoregressive prior [? ? ?]. Two-stage designs fit where counts allow, pool the reduced curves in a multivariate meta-regression on area-level vulnerability predictors, and predict every area from its own profile [? ? ? ?]. A smoothing prior pools neighbours hardest in the temperature tail, where vulnerability binds and heat risk is read; the two-stage design carries no geographic term, so each area’s risk follows its own profile.

We apply the two-stage small-area design [?] to the [XX] dissemination areas of Canada’s 41 census metropolitan areas: a case time series fit within each city and age band, pooled across cities on composite vulnerability components, and downscaled to an exposure-response curve for every dissemination area. Doing so asks two questions of the Canadian substrate: to what extent heat-related mortality risk varies across neighbourhoods, and how much of that variation the vulnerability indicators account for.

2 Methods

2.1 Study design and data

We performed a nationwide small-area analysis of mortality and temperature across the [XX] dissemination areas (DAs) within Canada’s 41 census metropolitan areas (CMAs), from 2000 to 2023.

Individual death records from the Canadian Vital Statistics Death database (CVSD), obtained through Statistics Canada’s virtual Data Lab, were assigned to DAs by residential postal code through the PCCF+ linkage and aggregated to daily counts of all-cause non-accidental deaths in four age bands (0–64, 65–74, 75–84, 85 and over). Each DA carries a daily mean temperature series, the mean of daily minimum and maximum from the Daymet V4 1 km gridded product,

area-weighted over its polygon. Age-specific population counts come from the 2021 Census, on 2021 boundaries throughout. DAs with zero population or without temperature coverage were excluded.

Area-level vulnerability indicators come from thirteen datasets of the Canadian Urban Environmental Health Research Consortium (CANUE; greenness, air quality, urban heat, built form, 2019), indexed to single-link postal codes at a 100 m buffer and assigned to DAs through the same PCCF+ linkage, and from thirteen topics of the 2021 Census Profile (demography, housing, deprivation), taken as rates. Air conditioning is not collected; renter share stands in for it. The Pampalon Material and Social Deprivation Index benchmarks the components and is not an input, as it is built from census variables that enter directly.

2.2 Statistical analysis

Daily death and temperature series enter as a case time series per DA [? ?], fitted separately for each CMA and age band, giving 164 first-stage estimates.

Each fit is a conditional Poisson model with quasi-Poisson variance, with stratum intercepts eliminated by conditioning [?]. Strata are defined by DA, year, and month, with day-of-week indicators and daily relative humidity [source] as covariates and no further smooth of time. Temperature enters through a cross-basis [?]: a quadratic B-spline with three internal knots at the 10th, 75th, and 90th percentiles of the CMA's temperature distribution, and a natural cubic spline over lags 0 to 21 with three internal knots equally spaced on the log scale. The two margins give five basis columns each, and the cross-basis 25 parameters.

We reduced each fitted cross-basis to its overall cumulative exposure-response curve, centred at the CMA's median temperature, yielding a five-coefficient summary and its covariance, the input to the second stage [?]. First-stage fits and reductions ran inside the virtual Data Lab; only the reduced coefficients and their covariances left it.

Vulnerability enters through a single principal component analysis fitted on the DA rows of all 41 CMAs; the first three components are carried forward, and CMA-level scores are population-weighted averages of the DA scores.

We pooled the 164 reduced curves in a multivariate meta-regression under restricted maximum likelihood, with age band indicators and the first three components as meta-predictors and a random intercept on CMA [? ? ?]. Residual heterogeneity is summarized by Cochran's Q and I^2 .

From the pooled model we predicted a curve for every DA and age band from the DA's own component scores, in two passes: a first prediction at the CMA-median centring locates the DA's minimum-mortality temperature, and the curve re-centred there yields the relative risks of heat and cold at the 99th and 1st percentiles and the corresponding attributable fractions [?]. Excess

mortality rates are standardized to the Canadian 2011 standard population. Empirical confidence intervals come from 1,000 Monte Carlo draws of the pooled coefficients from their estimated covariance, with the per-DA prediction repeated on each draw, CMA and national totals summed within each draw, and intervals taken at the 2.5th and 97.5th quantiles [?].

Sensitivity analyses varied the lag window (0–14, 0–28 days), the cross-basis dimension (9, 16), the exposure metric (daily maximum; apparent temperature), the humidity adjustment (excluded), the stratification (day of week absorbed into the strata), the time control (a natural cubic spline of time added), the season (warm season only), the CANUE buffer (postal code point value), the standard population (European Standard Population 2013), and the map spread (corrected for residual between-area variance). between-area variance). Secondary analyses repeated the model by cause of death and by sex. Analyses were performed in R [version] with the `dlm`, `gnm`, and `mixmeta` packages.

3 Results

The analysis included [XX] deaths from all causes across the 41 census metropolitan areas, spanning [XX] dissemination areas. Each year over the study period, an average of [XX] excess deaths (95% empirical CI [XX–XX]) were attributable to heat, corresponding to a standardized excess mortality rate of [XX] deaths per 100,000 person-years (Canadian 2011 standard population). Figure 1 shows the city-level heat relative risk at the 99th temperature percentile across the 41 CMAs, ranging from [XX] in [CMA] to [XX] in [CMA].

Figure 1: City-level relative risk of heat at the 99th temperature percentile across the 41 census metropolitan areas, from the first-stage fits. [Placeholder—gated on observed mortality.]

Risk increased with age. The pooled relative risk at the 99th percentile relative to the minimum-mortality temperature rose from [XX] (95% empirical CI [XX–XX]) at ages 0–64 to [XX] ([XX–XX]) at ages 85 and over (Figure 2). Minimum-mortality temperatures ranged from [XX] to [XX] °C across CMAs, tracking each city’s warm-season temperature distribution. Residual between-CMA heterogeneity after meta-regression was [Q p-value], I^2 [XX]%.

Figure 2: Pooled overall cumulative exposure-response curves by age band, from the multivariate meta-regression across the 41 CMAs. [Placeholder—gated on observed mortality.]

The first three components accounted for [XX]% of the variance of the indicators and entered the pooled model as meta-predictors, with [which components] associated with the heat slope;

[component] tracked the Pampalon material deprivation index ($r = [XX]$). Predicted curves for a high-vulnerability and a low-vulnerability profile within the same CMA differed by $[XX]\%$ in relative risk at the 99th percentile (Figure 3).

Figure 3: Vulnerability structure and its effect on risk: principal component loadings of the area-level indicators, and predicted exposure-response curves for high- and low-vulnerability profiles within one CMA. [Placeholder—gated on observed mortality.]

Figure 4 maps heat-related mortality risk at dissemination-area resolution nationally. Predicted relative risk at the 99th percentile ranged from $[XX]$ to $[XX]$ across DAs, and minimum-mortality temperatures from $[XX]$ to $[XX]^\circ\text{C}$. Variation within CMAs [exceeded / compared to] variation between them, with the highest risks concentrated in [spatial pattern]. Downscaled predictions agreed with unpooled census-tract reference fits within $[XX]$ (appendix).

Figure 4: Heat-related mortality risk across Canadian dissemination areas: predicted relative risk of heat at the 99th temperature percentile, from the pooled model and each DA’s own vulnerability profile. [Placeholder—gated on observed mortality.]

4 Discussion

Across the dissemination areas of Canada’s 41 census metropolitan areas, heat accounted for an average of $[XX]$ excess deaths ($[95\% \text{ eCI}]$) each year, a standardized excess mortality rate of $[XX]$ per 100,000 person-years, with minimum-mortality temperatures ranging from $[XX]$ to $[XX]^\circ\text{C}$ across cities. Risk varied [more within cities than between them], with the highest values concentrated in [pattern].

The patterns align with the small-area literature where it exists. The component tracking material deprivation carried [higher] heat risk, consistent with small-area findings in Europe $[? ?]$. [Density, green space, and air quality loadings] recur across cities with only their weights differing $[?]$. Minimum-mortality temperatures tracked each city’s own distribution, [continental CMAs separating from maritime ones], the adaptation signature reported across countries $[?]$. [Renter share], the available proxy for air conditioning, [behaved as in] the Canadian multi-city record $[?]$.

A map at this resolution places risk at the scale where heat-health response is organized. Public health units can identify the most exposed neighbourhoods directly rather than infer them from city-wide averages, and the fitted risk functions couple to weather forecasts for anticipatory warning $[?]$. The same framework extends to hospital admissions and emergency-department visits,

and to projection under climate scenarios. We recommend the dissemination-area estimates, not city averages, as the input layer for municipal heat-response planning.

The magnitude of the excess depends on the effect summary. Each estimate is referenced to the DA's minimum-mortality temperature, so the excess counts deaths against a counterfactual at that DA's own optimum; heat and cold burdens are therefore not comparable across studies using fixed thresholds. The estimates are averages over the study period and carry no temporal change in susceptibility.

This analysis has limitations. First, the downscale assumes that risk patterns estimated across areas hold at the areas themselves; agreement with unpooled census-tract fits (appendix) supports the assumption at one scale below the fitting unit. Second, the composite components size how much variation the indicators track and cannot attribute it to any single driver. Third, individual-level factors are not modelled, and the associations retain an ecological character. In addition, exposure comes from an interpolated grid whose uncertainty is not propagated, and predicted curves carry only the between-area variation the components explain, so the ordering of areas is more reliable than the range of risk; a surface corrected for the omitted variance taken as rates is reported as a sensitivity analysis.

References